In this Learning Module you will learn how to find all of the **factors** of individual whole numbers and the **common factors** of pairs or sets of whole numbers. In order to do this, you will first need to understand what **factors** are. Read the definition below.

Factor

In general, a factor is a **number that is multiplied by another number** in a multiplication situation resulting in a product.

For example, in the equation 3 x 8 = 24, the numbers **3** and **8** are **factors**.

Because 3 and 8 are **factors** of 24, this means that the number 24 can also be evenly divided by both 3 and 8.

Go back and reread the definition above, using your marker to circle the words in the definition that tell you what a **factor** is.

In the problems below, the factors have been circled. Use your marker to trace the circles around the factors.

$$\begin{array}{r} 5 \\ \times\ 8 \\ \hline 40 \end{array} \qquad 7 \times 6 = 42 \qquad \begin{array}{r} 4 \\ \times\ 9 \\ \hline 36 \end{array}$$

Use your marker to write the word **factor** in each box below.

factor

8 x 7 = 56

factor

factor

9 x 6 = 54

factor

factor

→12

x 6 ←

72

factor

factor

factor

2 x 3 x 4 = 24

factor

Factors

Read the information and follow the instructions.

Use your marker to trace the **factors** in the equations below.

3 x 7 = 21

6 x 8 = 48

12 x 2 = 24

3 x 4 = 12

4 x 8 = 32

18 x 2 = 36

6 x 2 = 12

8 x 7 = 56

4 x 2 = 8

2 x 8 = 16

9 x 7 = 63

5 x 5 = 25

Practice Identifying the Factors of a Number

Read the information and follow the instructions.

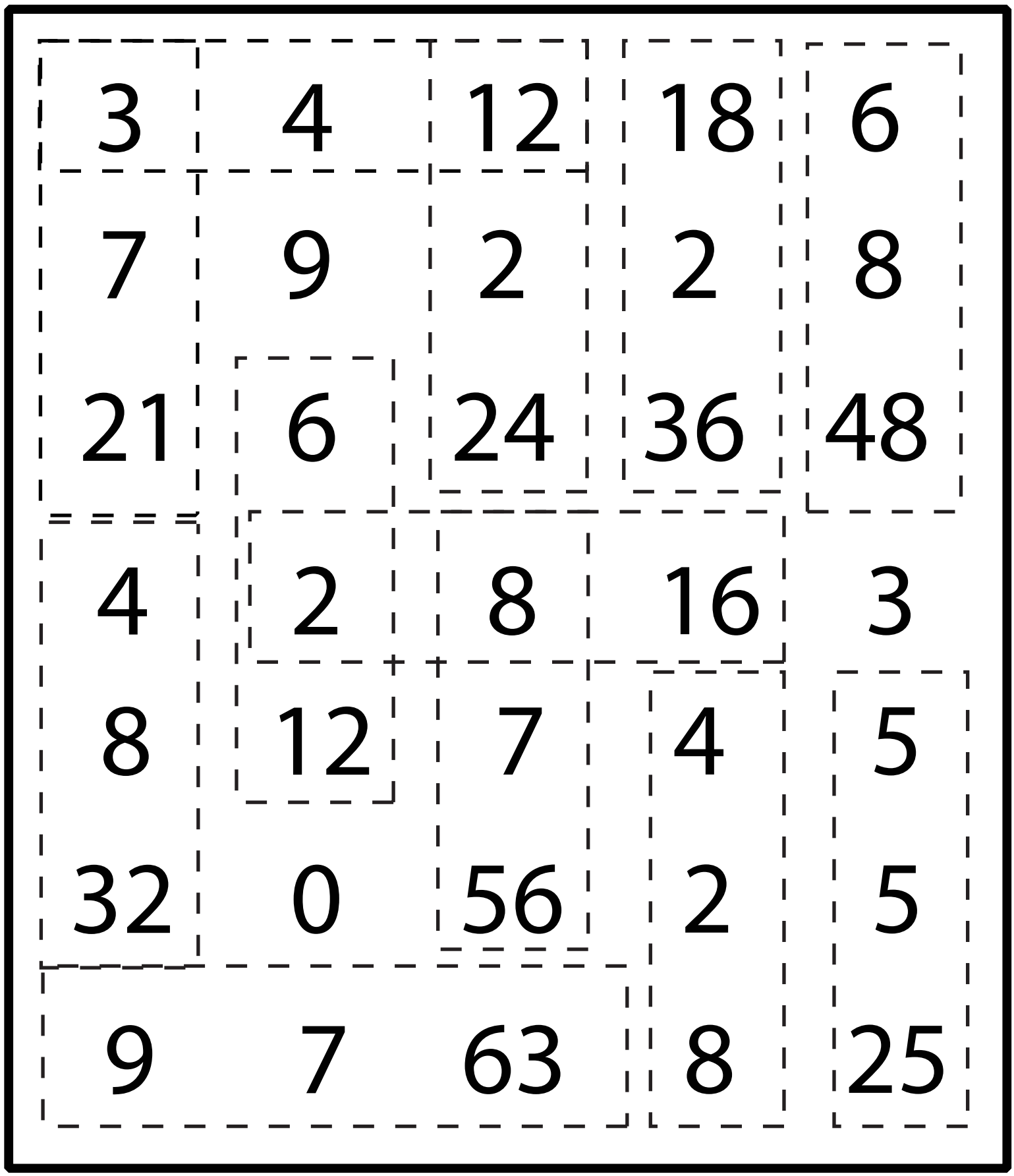
Box the **factors** and products in the rectangle below. The boxes may be horizontal or vertical. The first one has been done for you.

3	4	12	18	6
7	9	2	2	8
21	6	24	36	48
4	2	8	16	3
8	12	7	4	5
32	0	56	2	5
9	7	63	8	25

Factor Find

Read the information and follow the instructions.

Check your work. Did you find all of the sets of **factors** and products? If not, you can go back to Q 4 and try it again.



Now that you understand what **factors** are, you will be able to use that knowledge to find all of the f**actors** of given whole numbers. Think about the number 12 as the product (or the answer in a multiplication equation) of two **factors**. The list below shows all of the different **2-factor** equations having the number 12 as their product.

1 x 12 = 12

4 x 3 = 12

2 x 6 = 12

6 x 2 = 12

3 x 4 = 12

12 x 1 = 12

Notice that the numbers 1, 2, 3, 4, 6, and 12 are all **factors** of the number 12.

Use your marker to circle all of the **factors** of 12 in the above equations.

The list below shows all of the **factors** for the number 18.

1 x 18 = 18	6 x 3 = 18
2 x 9 = 18	9 x 2 = 18
3 x 6 = 18	18 x 1 = 18

Notice that the **factors** 1, 2, 3, 6, 9, and 18 in the last three equations have been written in **bold**. This is because they repeat the factors from the first 3 equations. When finding all of the **factors** of a number, it will only be necessary to list each factor **ONCE**. The **factors** of a number are usually written in ascending order (least to greatest).

What are all of the **factors** of 18?

1, 2, 3, 6, 9, 18

Using your marker, underline the question above and trace each of the **factors** in the set of numbers that answers that question.

Look at the sets of equations below. Use your marker to trace the **factors** written in the **boxes**, then write **all** of the **factors** for each product on the lines below the equation sets.

Set A

$\boxed{1} \times \boxed{28} = 28$

$\boxed{2} \times \boxed{14} = 28$

$\boxed{4} \times \boxed{7} = 28$

$7 \times 4 = 28$

$14 \times 2 = 28$

$28 \times 1 = 28$

What are all the factors of 28?

_____

Set B

$\boxed{1} \times \boxed{32} = 32$

$\boxed{2} \times \boxed{16} = 32$

$\boxed{4} \times \boxed{8} = 32$

$8 \times 4 = 32$

$16 \times 2 = 32$

$32 \times 1 = 32$

What are all the factors of 32?

_____

Solutions

1 2 4 7 14 28 1 2 4 8 16 32

Finding All Of The Factors

Read the information and follow the instructions.

Look at the sets of equations below. Use your marker to write the **factors** that belong in each of the empty **boxes**, then write **all** of the **factors** for each product on the lines below the equation sets.

Set A	Set B
☐ x ☐ = 44	☐ x ☐ = 50
☐ x ☐ = 44	☐ x ☐ = 50
☐ x ☐ = 44	☐ x ☐ = 50
11 x 4 = 44	10 x 5 = 50
22 x 2 = 44	25 x 2 = 50
44 x 1 = 44	50 x 1 = 50

What are all the factors of 44? _____

What are all the factors of 50? _____

Look at the sets of equations below. Use your marker to write the **factors** that belong in each of the empty **boxes**, then write **all** of the **factors** for each product on the lines below the equation sets.

Set A

$\square \times \square = 44$

1 44

$\square \times \square = 44$

2 22

$\square \times \square = 44$

4 11

$11 \times 4 = 44$

$22 \times 2 = 44$

$44 \times 1 = 44$

Set B

$\square \times \square = 50$

1 50

$\square \times \square = 50$

2 25

$\square \times \square = 50$

5 10

$10 \times 5 = 50$

$25 \times 2 = 50$

$50 \times 1 = 50$

What are all the factors of 44?

_____

1, 2, 4, 11, 22, 44

What are all the factors of 50?

_____

1, 2, 5, 10, 25, 50

When finding all of the **factors** of a whole number it is not always necessary to list every 2-factor equation having that whole number as their product. As you have learned, there comes a point when the **factors** start to repeat. Using a simple method of writing the **factors** in pairs following a specific pattern, you can quickly make a list of all of the **factors** of any whole number. Here's how it works:

What are all of the factors of 24? _____

1) Write the number 1-10 above the question.

1 2 3 4 5 6 7 8 9 10

What are all of the factors of 24? _____

2) The number 1 is a factor of every number. Put a check mark above the number 1 and record the 1 and 24 (the factor pair) in the factor list.

?
1 2 3 4 5 6 7 8 9 10

What are all of the factors of 24? 1 _____ 24

Use your marker to circle the numbers 1-10 in the first step of the example above. Next, trace the check mark above the number 1 in the second step of the example and then write a 1 and a 24 above these numbers in the factor list.

3) Next, try the number 2 from the 1-10 list. The number 2 will be a **factor** for any **even number.** Since 24 is even, 2 is a factor of 24. Think, "What times 2 is 24?" Since 2 x 12 = 24, put a check mark above the number 2 and record the 2 and the 12 (**the factor pair**) in the **factor list**.

? ?
1 2 3 4 5 6 7 8 9 10

What are all of the factors of 24? 1 2 12 24

Use your marker to trace the check marks above the numbers 1 and 2 in the third step of the example and then write a 1, 2, 12, and 24 above these numbers in the **factor list**.

4) Next, try the number 3 from the 1-10 list. Since 24 will divide evenly by 3, 3 is a **factor** of 24. Think, "What times 3 is 24?" Since 3 x 8 = 24, put check marks above the numbers 3 and 8 and record the 3 and 8 (**the factor pair**) in the **factor list**.

? ? ? ?
1 2 3 4 5 6 7 8 9 10

What are all of the factors of 24? 1 2 3 8 12 24

Use your marker to trace the check marks above the numbers 1, 2, 3, and 8 in the fourth step of the example and then write a 1, 2, 3, 8,12, and 24 above these numbers in the **factor list**.

5) Next, try the number 4 from the 1-10 list. Since 24 will divide evenly by 4, 4 is a **factor** of 24. Think, "What times 4 is 24?" Since 4 x 6 = 24, put check marks above the numbers 4 and 6 and record the 4 and 6 (the **factor pair**) in the **factor list**.

? ? ? ? ? ?
1 2 3 4 5 6 7 8 9 10

What are all of the factors of 24? 1 2 3 4 6 8 12 24

Use your marker to trace the check marks above the numbers 1, 2, 3, 4, 6, and 8 in the fifth step of the example and then write a 1, 2, 3, 4, 6, 8,12, and 24 above these numbers in the **factor list**.

6) Next, try the number 5 from the 1-10 list. Since 24 will NOT divide evenly by 5, 5 is NOT a factor of 24. Put an X above the number 5.

? ? ? ? X ? ?
1 2 3 4 5 6 7 8 9 10

What are all of the factors of 24? 1 2 3 4 6 8 12 24

Use your marker to trace the check marks and the X above the numbers in the list.

7) Next, try the number 7 from the 1-10 list. Since 24 will **NOT** divide evenly by 7, 7 is **NOT** a factor of 24. Put an X above the number 7.

? ? ? ? X ? X ?
1 2 3 4 5 6 7 8 9 10

What are all of the factors of 24? 1 2 3 4 6 8 12 24

Use your marker to trace the check marks and the X above the numbers in the list.

8) Next, try the number 9 from the 1-10 list. Since 24 will **NOT** divide evenly by 9, 9 is **NOT** a factor of 24. Put an X above the number 9.

? ? ? ? X ? X ? X
1 2 3 4 5 6 7 8 9 10

What are all of the factors of 24? 1 2 3 4 6 8 12 24

Use your marker to trace the check marks and the X above the numbers in the list.

9) Next, try the number 10 from the 1-10 list. Since 24 will **NOT** divide evenly by 10, 10 is **NOT** a factor of 24. Put an X above the number 10.

? ? ? ? X ? X ? X X
1 2 3 4 5 6 7 8 9 10

What are all of the factors of 24? 1 2 3 4 6 8 12 24

Use your marker to trace the check marks and the X above the numbers in the list. You now have a complete list of all of the **factors** of 24!

1) Write the numbers 1-10 above the question below.

What are all of the factors of 28? _____

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2) The number 1 is a factor of every number. Put a check mark above the number 1 and record the 1 and the 28 (the **factor pair**) in the **factor list**.

1 2 3 4 5 6 7 8 9 10

What are all of the factors of 28? _____

---

3) Next, try the number 2 from the 1-10 list. Since 28 is even, 2 is a factor of 28. Think, "What times 2 is 28?" Since 2 x 14 = 28, put a check mark above the number 2 and record the 2 and the 14 (the **factor pair**) in the **factor list**.

? 1 2 3 4 5 6 7 8 9 10

What are all of the factors of 28? 1 _____ 28

---

4) Next, try the number 3. Since 28 will NOT divide evenly by 3, put an X above the number 3.

? ? 1 2 3 4 5 6 7 8 9 10

What are all of the factors of 28? 1 2 _____ 14 28

5) Next, try the number 4 from the list. Since 28 will divide evenly by 4, think, "What times 4 is 28?" Since 4 x 7 = 28, put check marks above the numbers 4 and 7 and record the 4 and 7 (the **factor pair**) in the **factor list**.

? ? X
1 2 3 4 5 6 7 8 9 10

What are all of the factors of 28? 1 2 14 28

---

6) Next, try the number 5. Since 28 will **NOT** divide evenly by 5, put an X above the number 5.

? ? X ? ?
1 2 3 4 5 6 7 8 9 10

What are all of the factors of 24? 1 2 4 7 14 28

---

7) Next, try the number 6. Since 28 will **NOT** divide evenly by 6, put an X above the number 6.

? ? X ? X ?
1 2 3 4 5 6 7 8 9 10

What are all of the factors of 24? 1 2 4 7 14 28

---

8) Next, try the numbers 8, 9, and 10. Since 28 will **NOT** divide evenly by 8, 9 or 10, put an X above each of these numbers.

? ? X ? X X ?
1 2 3 4 5 6 7 8 9 10

What are all of the factors of 24? 1 2 4 7 14 28

Use the 1-10 list method to find all of the **factors** for the following numbers.

1) What are all of the factors of 21? _____

2) What are all of the factors of 27? _____

3) What are all of the factors of 30? _____

4) What are all of the factors of 36? _____

Solutions

1) 1 3 7 21

2) 1 3 9 27

3) 1 2 3 5 6 10 15 30

4) 1 2 3 4 6 9 12 18 36

Notice that the factor 6 in the last problem is circled. When checking off the 1-10 list, the number 36 divides evenly by 6. When you think "What times 6 is 36?" you discover that 6 x 6 = 36. Remember that we only need to list any factor ONCE. Since 6 would be its own partner, circle it.

Use the 1-10 list method to find all of the **factors** for the following numbers.

1) What are all of the factors of 15? _____

2) What are all of the factors of 35? _____

3) What are all of the factors of 52? _____

4) What are all of the factors of 16? _____

5) What are all of the factors of 25? _____

Check Your Factor Skills *Follow the directions.*

Check your work. Were you able to find all of the **factors** for the numbers? If not, go back to Q 18 and try again.

1) What are all of the factors of 15?

1 3 5 15

? x ? x ? x x x x x
1 2 3 4 5 6 7 8 9 10

2) What are all of the factors of 35?

1 5 7 35

? x x x ? x ? x x x
1 2 3 4 5 6 7 8 9 10

3) What are all of the factors of 52?

1 2 4 13 26 52

? ? x ? x x x x x x
1 2 3 4 5 6 7 8 9 10

4) What are all of the factors of 16?

1 2 4 8 16

? ? x ? x x x ? x x
1 2 3 4 5 6 7 8 9 10

5) What are all of the factors of 25?

1 5 25

? x x x ? x x x x x
1 2 3 4 5 6 7 8 9 10

ACROSS

1 List all of the factors of 48

2 List all of the factors of 2

3 List all of the factors of 18

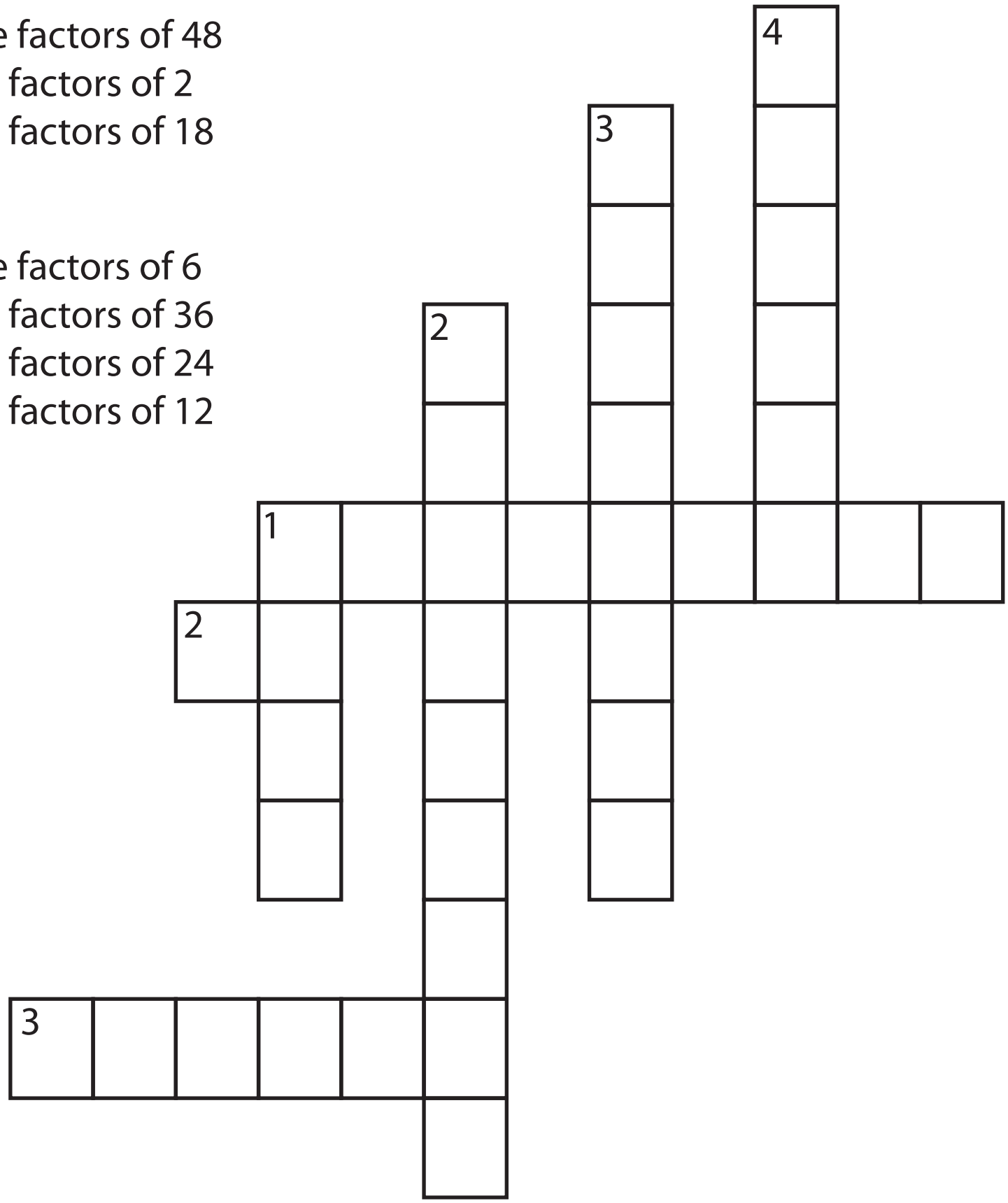
DOWN

1 List all of the factors of 6

2 List all of the factors of 36

3 List all of the factors of 24

4 List all of the factors of 12



ACROSS	DOWN
1) 1 2 3 4 6 8 12 16 24 48	1) 1 2 3 6
2) 1 2	2) 1 2 3 4 6 9 12 18 36
3) 1 2 3 6 9 18	3) 1 2 3 4 6 8 12 24
	4) 1 2 3 4 6 12

Cross-Factor Puzzle

Fill in the boxes

You have learned how to find all of the factors of individual whole numbers. The rest of this Module will teach you how to find the **common factors** for pairs or sets of whole numbers. Start by reading the definition below.

common factor

a number that is a **factor** of **two or more** whole numbers.

Use your marker to underline the term **common factor above**. Then underline the words in the definition that are the **most important** to remembering what a common factors is.

Rewrite the definition of a **common factor** on the lines below;

common factor _____

_____

To find the **common factors** of a pair of whole numbers, it is first necessary to find all of the factors for each of the numbers.

What are the common factors of 12 and 18?
List the factors for 12 and 18 using the 1-10 list method.

? ? ? ? x ? x x x x
1 2 3 4 5 6 7 8 9 10

12 1 2 3 4 6 12

? ? ? x x ? x x x x
1 2 3 4 5 6 7 8 9 10

18 1 2 3 6 9 18

Circle the factors that appear in **BOTH** lists.
The circled numbers are the common factors of 12 and 18.

? ? ? ? x ? x x x x
1 2 3 4 5 6 7 8 9 10

12 1 2 3 4 6 12

? ? ? x x ? x x x x
1 2 3 4 5 6 7 8 9 10

18 1 2 3 6 9 18

What are the common factors of 12 and 18? 1 2 3

Use your marker to complete the exercises below.

1) What are the common factors of 16 and 24? _____

1 2 3 4 5 6 7 8 9 10

16 _____

1 2 3 4 5 6 7 8 9 10

24 _____

2) What are the common factors of 18 and 30? _____

1 2 3 4 5 6 7 8 9 10

18 _____

1 2 3 4 5 6 7 8 9 10

30 _____

Solutions 1) 1 2 4 8 2) 1 2 3 6

Practicing Finding Common Factors

Read the information and follow the instructions.

Find all of the **common factors** for the number pairs below using the method you have learned.

1) What are the common factors of 15 and 30? _____

2) What are the common factors of 12 and 32? _____

3) What are the common factors of 20 and 40? _____

Practicing Finding Common Factors

Follow the directions.

Check your work. Were you able to find the **common factors** for the number pairs? If not, go back to Q 24 and try again.

1) What are the common factors of 15 and 30? _____

1 3 5 15

2) What are the common factors of 12 and 32? _____

1 2 4

3) What are the common factors of 20 and 40? _____

1 2 4 5 10 20

Find the **common factors** for the number sets below.

1) What are the common factors of 15, 30, and 45? _____

2) What are the common factors of 12, 28, and 32? _____

Finding Common Factors for Number Sets

Follow the directions.

Check your work. Were you able to find the **common factors** for the number sets? If not, go back to Q 26 and try again.

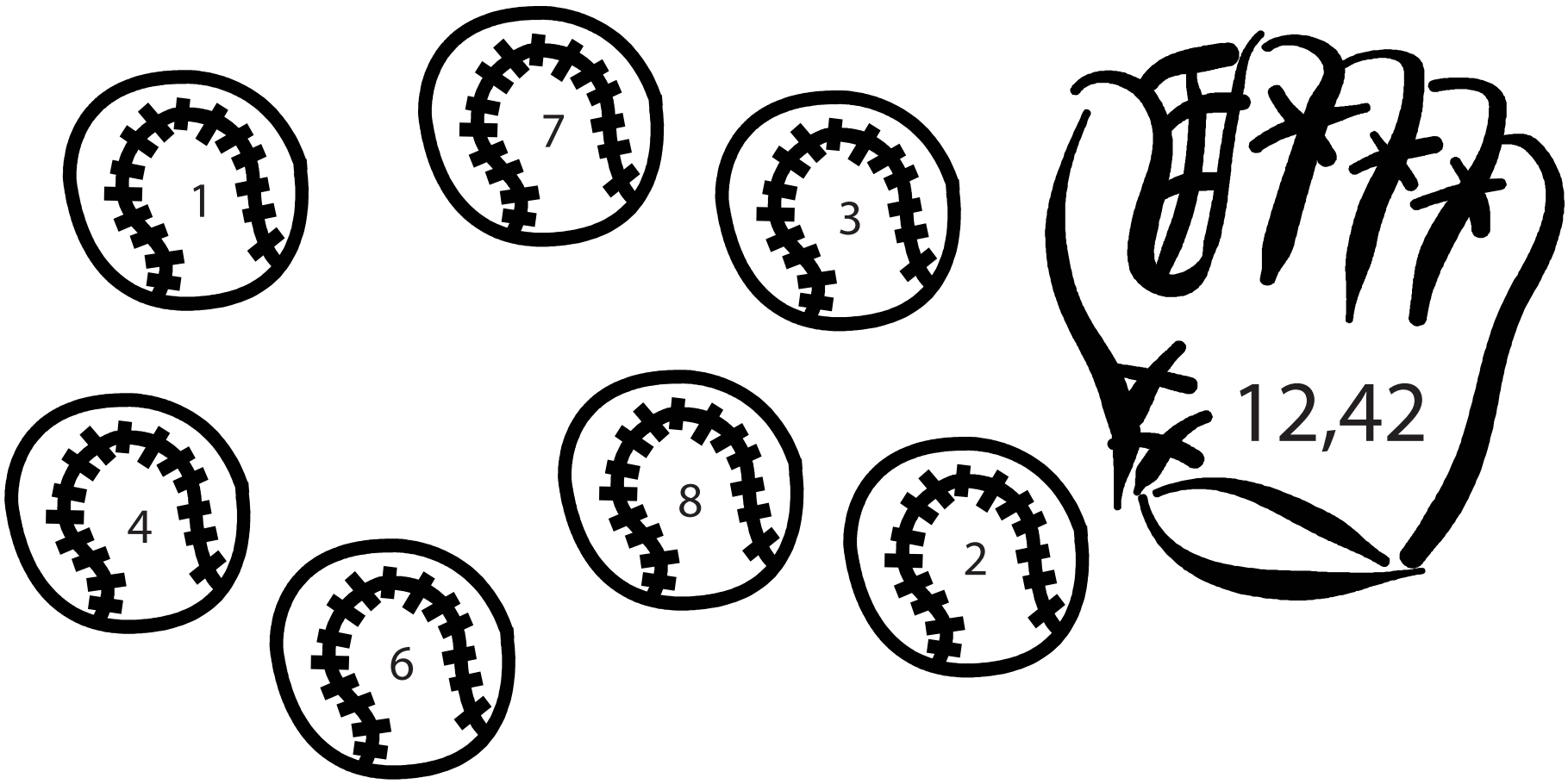
1) What are the common factors of 15, 30, and 45?

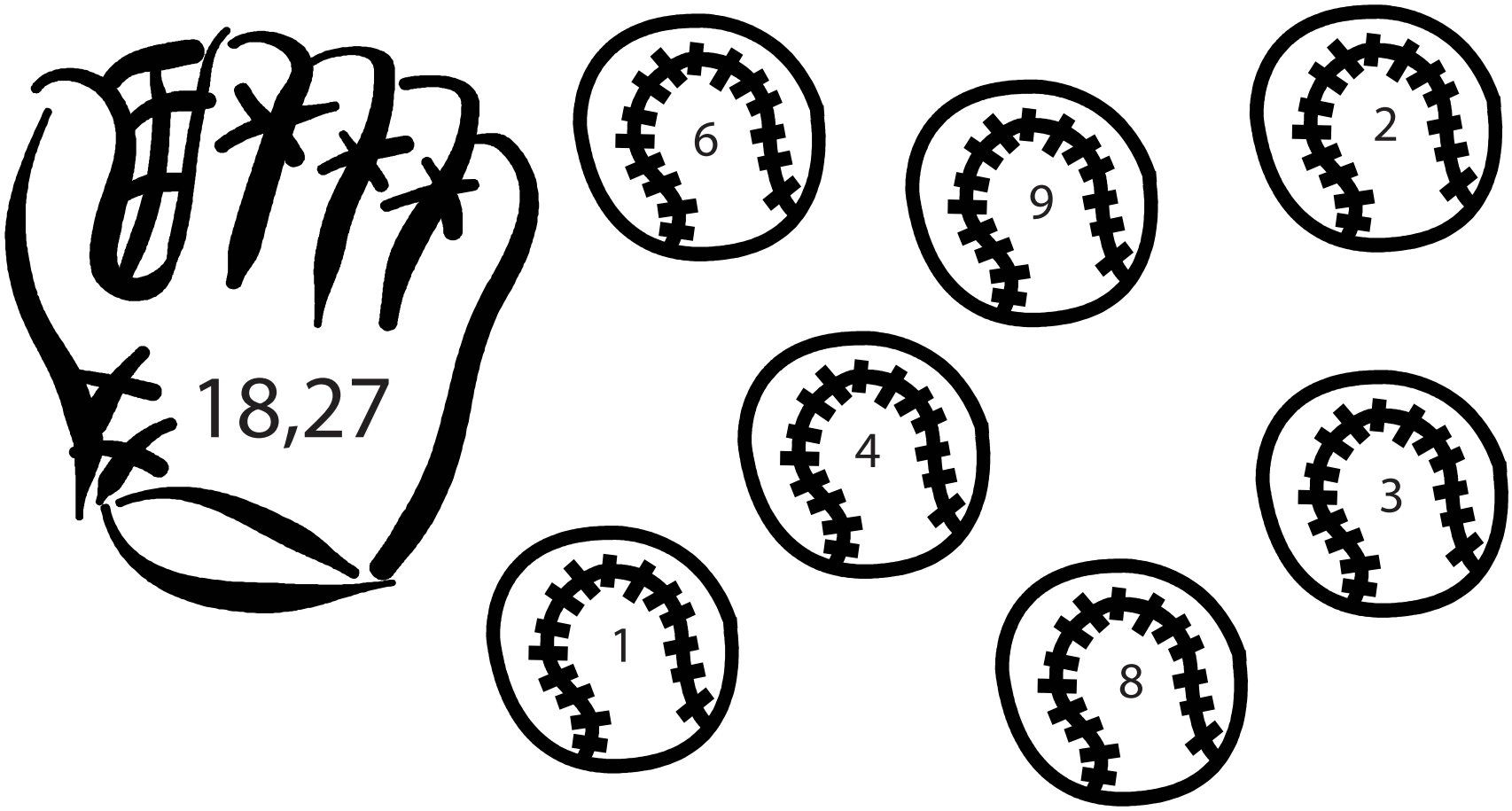
1 3 5 15

2) What are the common factors of 12, 28, and 32?

1 2 4

Look at the number pairs in each baseball glove. Catch (**by circling**) the baseballs beside each glove that contain the **common factors** for the numbers in the glove.





Solutions

12, 42 1 3 6 18, 27 1 3 9

Catching Common Factors *Follow the directions.*

Sometimes you may have to answer multiple choice questions involving **factors** or **common factors**. The important thing to remember is that each answer choice is there for a reason. The best strategy for choosing the correct answer is to **do your own work first** and then **find the answer choice** that matches your work!

1) Which of the following lists all of the factors of 24?

A 1 , 2 , 3 , 8 , 12 , 24

B 1 , 2 , 3 , 4 , 8 , 12 , 24

C 2 , 3 , 4 , 6 , 8 , 12 , 24

D 1 , 2 , 3 , 4 , 6 , 8 , 12 , 24

Find the factors of 24 using the 1-10 list method.

?	?	?	?	x	?	x	?	x	x
1	2	3	4	5	6	7	8	9	10

Pick the answer choice that matches your work!

Circle the correct answer choice for problem 1 then read and circle the correct answer for number 2 below.

2) Which of the following is **NOT** a factor of 63?

A 9 C 6

B 7 D 3

Find the factors of 63 using the 1-10 list method.

?	x	?	x	x	x	?	x	?	x
1	2	3	4	5	6	7	8	9	10

Pick the answer choice that matches your work!

Circle the letter for the correct answer to each question.

1) Which of the following lists all of the **factors** of 54?

A 1 , 2 , 3 , 8 , 12 , 24

B 1 , 2 , 3 , 4 , 8 , 12 , 24

C 2 , 3 , 4 , 6 , 8 , 12 , 24

D 1 , 2 , 3 , 4 , 6 , 8 , 12 , 24

2) Which of the following is **NOT** a factor of 63?

A 9 C 6

B 7 D 3

3) Which of the following lists all of the **common factors** of 27 and 36?

A 1 , 3 , 6

B 1 , 3 , 9

C 1 , 2 , 3 , 9

D 1 , 2 , 3 , 6 , 9

1) D 2) C 3) B

It's time to show how much you have learned about **factors** and **common factors**. Circle the letter for the correct answer to each question.

1) Which of the following lists all of the factors of 40?

A 1 , 2 , 5 , 8 , 20 , 40

B 1 , 2 , 4 , 8 , 10 , 20 , 40

C 1 , 2 , 4 , 5 , 8 , 10 , 20 , 40

D 1 , 2 , 4 , 5 , 8 , 10 , 40

2) Which of the following is **NOT** a factor of 56?

A 2 C 7

B 6 D 8

3) Which of the following lists all of the common factors of 24 and 30?

A 1 , 3 , 6

B 1 , 2 , 3 , 4

C 2 , 3 , 4 , 6

D 1 , 2 , 3 , 6

Check your work. Did you answer all of the questions correctly?
If not, go back to Q 31 and try again.

1) Which of the following lists all of the **factors** of 40?

A 1 , 2 , 5 , 8 , 20 , 40

B 1 , 2 , 4 , 8 , 10 , 20 , 40

→ C 1 , 2 , 4 , 5 , 8 , 10 , 20 , 40

D 1 , 2 , 4 , 5 , 8 , 10 , 40

2) Which of the following is **NOT** a factor of 56?

A 2 C 7

→ B 6 D 8

3) Which of the following lists all of the **common factors** of 24 and 30?

A 1 , 3 , 6

B 1 , 2 , 3 , 4

C 2 , 3 , 4 , 6

→ D 1 , 2 , 3 , 6

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